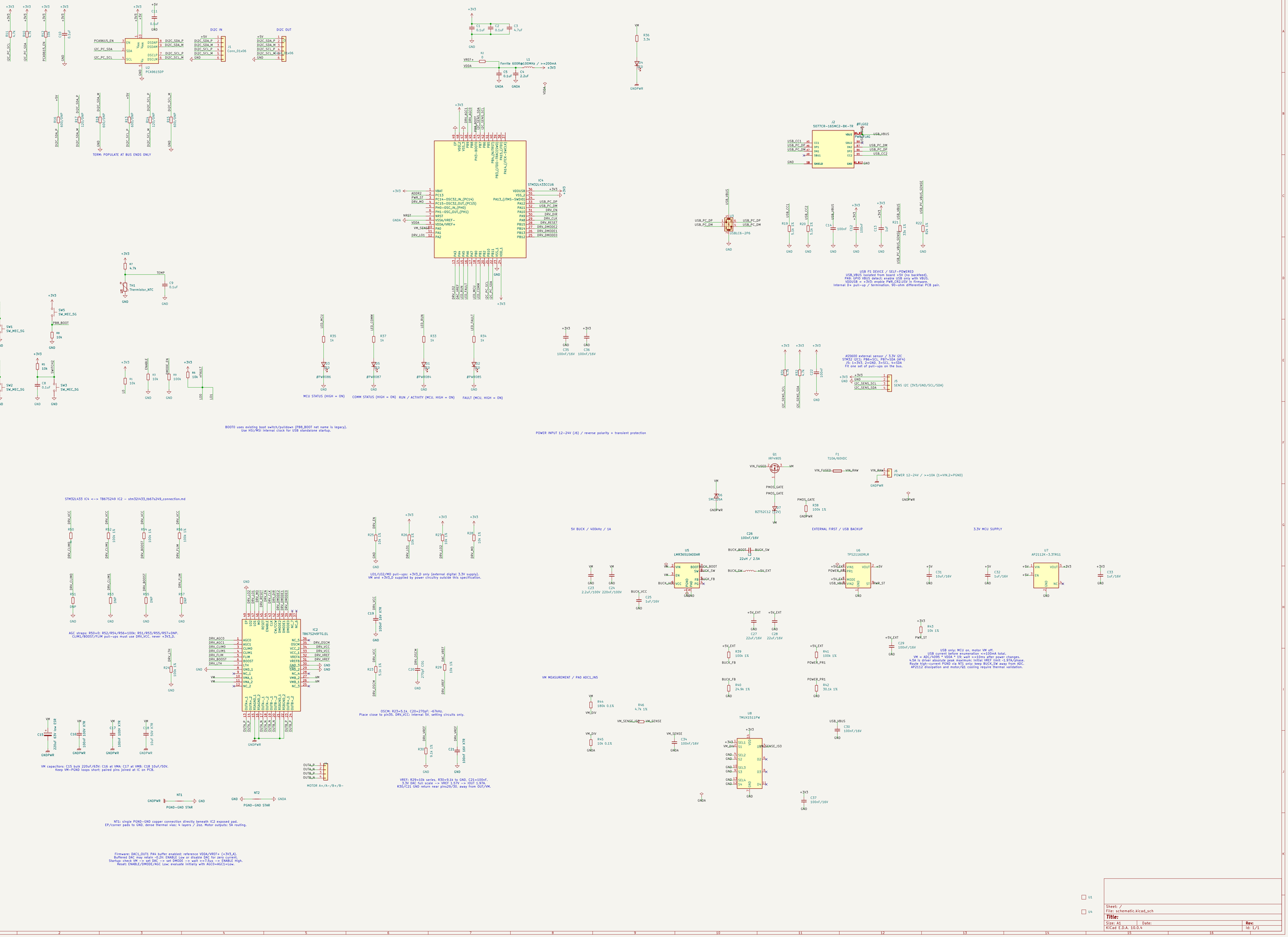

I2C_PC_SCL
I2C_PC_SDA
PCA9615_EN
PCA9615_INT
PCA9615 EN
I2C_PC_SCL
DI2C IN
DI2C OUT
PCA9615_EN
I2C_PC_SDA
I2C_PC_SCL
U2
PCA9615DP
DI2C_SDA_P
DI2C_SDA_M
DI2C_SCL_P
DI2C_SCL_M
J1
Conn_01x06
J2
TERM: POPULATE AT BUS ENDS ONLY
VREF+
VDDA
Ferrite 600R@100MHz / >=200mA
GNDA
VDDA
VM
R36
3.3k
D4
GNDPWR
IC4
STM32L433CCU6
VBAT
PC13
PC14-OSC32_IN_(PC14)
PC15-OSC32_OUT_(PC15)
PH0-OSC_IN_(PH0)
PH1-OSC_OUT_(PH1)
NRST
VSSA/VREF-
VDDA/VREF+
PA0
PA1
PA2
VDDUSB
VSS_2
PA13_(JTMS-SWDIO)
PA12
PA11
PA10
PA9
PA8
PB15
PB14
PB13
PB12
USB_PC_DP
USB_PC_DM
DRV_EN
DRV_DIR
DRV_CLK
DRV_RESET
DRV_DMODE2
DRV_DMODE1
DRV_DMODE0
ADDR2
PWR_ST
DRV_MO
VM_SENS
DRV_LO1
TEMP
TH1
Thermistor_NTC
SW1
SW_MEC_5G
SW5
PB8_BOOT
SW2
SW3
SWITCH1
SWITCH2
LED_MCU
LED_COMM
LED_RUN
LED_FAULT
MCU STATUS (HIGH = ON)
COMM STATUS (HIGH = ON)
RUN / ACTIVITY (MCU, HIGH = ON)
FAULT (MCU, HIGH = ON)
BOOT0 uses existing boot switch/pulldown (PB8_BOOT net name is legacy).
Use HSI/MSI internal clock for USB standalone startup.
POWER INPUT 12–24V (J6) / reverse polarity + transient protection
USB_VBUS
USB_PC_DP
USB_PC_DM
USB_CC1
USB_CC2
USB_PC_VBUS_SENSE
USB_PC_VBUS_SENSE
USB FS DEVICE / SELF-POWERED
USB_VBUS isolated from board +5V (no backfeed).
PA9_SPI0 VBUS detect; enable USB only with VBUS.
VDDUSB = +3V3; enable PWR_M2_SV in firmware.
Internal 3× pull-up / termination, 90-ohm differential PCB pair.
A55600 external sensor / 3.3V I2C
STM32 I2C1: PB6=SCL, PB7=SDA (AF4)
J5: 1=3V3, 2=GND, 3=SCL, 4=SDA
Fit one set of pull-ups on the bus.
SENS I2C (3V3/GND/SCL/SDA)
I2C_SENS_SCL
I2C_SENS_SDA
AGC straps: R50=0; R52/R54/R56=100k; R51/R53/R55/R57=DNP.
CLIM1/BOOST/FLIM pull-ups must use DRV_VCC, never +3V3_D.
STM32L433 IC4 <-> TB67S249 IC2 — stm32l433_tb67s249_connection.md
IC2
TB67S249FTG,EL
VM capacitors: C15 bulk 220uF/63V; C16 at VMA; C17 at VMB; C18 10uF/50V.
Keep VM–PGND loops short; paired pins joined at IC on PCB.
MOTOR A+/A−/B+/B−
NT1
PGND–GND STAR
NT2
PGND–GND STAR
NT1: single PGND–GND copper connection directly beneath IC2 exposed pad.
EP/corner pads to GND; dense thermal vias; 4 layers / 2oz. Motor outputs: 5A routing.
Firmware: DAC1_OUT1 PA4 buffer enabled; reference VDDA/VREF+ (+3V3_A).
Buffered DAC may retain ~0.2V. ENABLE Low or disable DAC for zero current.
Startup: check VM -> set DAC -> set DMODE -> wait >=7.5us -> ENABLE High.
Reset: ENABLE/DMODE/AGC Low; evaluate initially with AGC0=AGC1=Low.
OSCM: R23=5.1k, C20=270pF; ~67kHz.
Place close to pin35. DRV_VCC: internal 5V, setting circuits only.
VREF: R29=10k series, R30=9.1k to GND, C21=100nF.
3.5V DAC full scale -> VREF 1.579 -> IOUT 1.97A.
R30/C21 GND return near pins29/30, away from OUT/VM.
VM MEASUREMENT / PAD ADC1_IN5
5V BUCK / 400kHz / 1A
LO1/LO2/MO pull-ups: +3V3_D only (external digital 3.3V supply).
VM and +3V3_D supplied by power circuitry outside this specification.
Q1
IRF4905
F1
T10A/60VDC
PMOS_GATE
D7
BZT52C12 (12V)
R38
100k 1%
EXTERNAL FIRST / USB BACKUP
3.3V MCU SUPPLY
U5
LMR36510ADDAR
U6
TPS21160RLR
U7
AP2112K-3.3TRG1
C26
100nF/16V
L2
22uH / 2.5A
BUCK_BOOT
BUCK_SW
BUCK_FB
BUCK_VCC
C25
1uF/16V
C23
2.2uF/100V
C24
220nF/100V
C27
22uF/16V
C28
22uF/16V
C31
10uF/16V
C32
1uF/16V
C33
1uF/16V
R39
100k 1%
R40
24.9k 1%
R41
100k 1%
R42
30.1k 1%
R43
10k 1%
C29
100nF/16V
USB only: MCU on, motor VM off.
USB current before enumeration <=100mA total.
VM = ADC/4095 * VDDA * 19; wait >=10ms after power changes.
4.5A is driver absolute peak maximum; Initial VREF limit ~1.97A/phase.
Route high-current PGND via NT1 only; keep BUCK_SW away from ADC.
AP2112 dissipation and motor/Q1 cooling require thermal validation.
U8
TMUX1511PW
VM_SENSE_ISO
VM_SENSE
VM_DIV
R44
180k 0.1%
R45
10k 0.1%
R46
4.7k 1%
C34
100nF/16V
C30
100nF/16V
C37
100nF/16V
Sheet: /
File: schematic.kicad_sch
Title:
Size: A1
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/1